

Exploring Emotional Responses with Dynamic Difficulty Adjustment Adaptation in Immersive Virtual Reality Exergaming

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Abstract—Immersive Virtual Reality (IVR) exergaming presents a promising avenue to integrate physical exercise with engaging virtual experiences, potentially encouraging sustained physical activity. However, maintaining user motivation over extended periods poses a significant challenge. Recent research has introduced the Dynamic Difficulty Adjustment (DDA) mechanism, dynamically regulating exergame difficulty based on specific conditions to enhance user adaptation. While prior studies have predominantly focused on gaming performance to adjust difficulty, they often overlook the emotional impact on user motivation. This study investigates users' emotional responses to the game timer change (TC) as a DDA mechanism during IVR exergaming. Results indicate that subjects in the TC-implemented game displayed more neutral emotions, concomitant with improved gaming performance. Conversely, subjects in the game without TC exhibited a broader range of detected emotions (sad and happy), suggesting difficulties in adapting to in-game difficulty levels incongruent with their gaming abilities. Overall, this study establishes a foundation for future research in affective computing-based IVR exergaming, aiming to develop an intelligent autonomous DDA mechanism tailored to users' physical and mental conditions.

Index Terms—Immersive virtual reality, dynamic difficulty adjustment, affective computing

I. INTRODUCTION

Exergaming refers to digital games that necessitate physical movements for gameplay, creating an active gaming experience serving as a form of physical activity [1]. It has been shown to positively influence and motivate individuals to participate in physical activity [2]. With the rapid advancement

of immersive virtual reality (IVR) technology, exergaming has evolved from a basic fitness activity to a novel interactive experience [3]. IVR exergaming seamlessly integrates physical exercise with gameplay within an immersive virtual setting, effectively promoting physical activity among users [4], [5]. This approach proved practical for addressing sedentary issues in individuals [6]. Recent research suggested that the immersive nature of IVR exergaming significantly boosted the user's sense of energy and motivation during training, potentially leading to improved aerobic exercise performance [7], [8]. However, an ongoing challenge lies in enhancing user motivation to engage in IVR exergaming. This proves a challenging issue, as several factors including: user fitness levels; preferences; and objectives; must be considered. Consequently, designing adaptable game mechanics that cater to diverse users and sustain their engagement emerged as a pragmatic solution to address this issue.

Hence, a dynamic difficulty adjustment (DDA) mechanism was proposed by [9], which automatically adjusts the game's difficulty level by analyzing the user's performance. This approach proved effective in reducing difficulty levels for beginners, thereby preventing early frustration [10]. Conversely, for more gaming-experienced users, the difficulty level can be elevated to sustain a challenging experience [10]. DDA had been widely implemented in non-immersive VR (NIVR) exergaming to customize game parameters, such as in-game object speed, to accommodate the varying capabilities of individuals in rehabilitation programs [11], [12]. Its integration into IVR exergaming further enhances the personalized and engaging nature of the gaming experience [13]–[15]. Despite its widespread use, there was a paucity of research addressing the emotional behavioral changes of users upon the activation of DDA mechanism in IVR exergaming.

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To address this challenge, this work proposes a study on the emotional changes of users in response to the dynamically adjustable timer change (TC) feature, a prevalent DDA mechanism, within an IVR exergame. The contributions of this research are outlined as follows:

- Analyzed the emotional changes by examining facial expressions and physiological signals of users during gameplay. The analysis was validated through the evaluation of characteristics from the Intrinsic Motivation Inventory (IMI) [16] among users.
- Explored the correlation between user emotions across two IVR exergaming experiences: one with the TC feature implemented and the other without it.

II. RELATED WORK

A. Exergaming in XR

The promotion of physical activity through exergaming has primarily aimed at addressing sedentary behavior and combating obesity. Engagement in exergaming resulted in a significant increase in heart rate (HR) and energy expenditure compared to sedentary behavior, with little difference observed in comparison to traditional moderate-intensity physical activities [17]. Sween *et al.* [18] identified a robust correlation between exergaming and heightened physical activity levels, predominantly achieving moderate-intensity activity levels in alignment with the American College of Sports Medicine guidelines for health and fitness. Additionally, evidence suggested that longer durations of exergaming may be more beneficial in meeting these recommendations [18].

In a study investigating the impact of IVR exergaming on high-intensity interval training (HIIT), Farrow *et al.* [19] revealed that IVR exergaming significantly enhanced the enjoyment of HIIT sessions for untrained individuals. The research also emphasized that allowing participants to compete against their own previous performance effectively increased exercise intensity in VR-HIIT sessions. Additionally, increasing the resistance in the game, such as the “hard mode”, the highest difficulty level for the game, led to improved cycling performance during HIIT, without compromising participants’ enjoyment. These findings were supported by post-exercise questionnaires assessing intrinsic motivation, subjective vitality, and future exercise intentions. On the other hand, Huang *et al.* [20] explored the impact of IVR exergaming on the executive functions of older adults. The results indicated that participating in an IVR environment positively influenced specific cognitive tasks, with the sense of presence playing a mediating role in this enhancement. These findings underscore the utility and impact of IVR exergaming in enhancing cognitive performance and potentially preventing cognitive decline among users.

B. DDA in XR-based Exergaming

Two fundamental components of the DDA mechanism in XR-based exergaming include the activation conditions of DDA within the exergames and the implementation methods of the DDA adaptation mechanism in the exergames [21]. Typically, the activation conditions of DDA in exergaming

can be categorized into two approaches: performance-based adaptation and physiological-based adaptation.

Performance-based adaptation involves assessing users’ performance during gameplay to adjust the difficulty level of the exergame. This assessment may include factors such as the damage inflicted on virtual enemies or targeted objects [22], the areas and distances covered during the collection of virtual items [23], health point status [24], movement speed and accuracy [25], or accurate responses to fulfill mission tasks [26]. Physiological-based adaptation employs physiological indicators to trigger the DDA mechanism, analyzing factors such as brain activities [27], eye movements [28], changes in HR [29], or facial expressions [30] of users during gameplay. In many instances, these approaches are used to infer mental states, such as excitement, anxiety, or attention levels.

For instance, Stein *et al.* [31] contributed to DDA by developing a method that adjusts difficulty levels based on the excitement level computed from measured EEG signals. The study observed an increase in user excitement in the exergaming with the implemented DDA mechanism compared to exergaming without DDA. Similarly, Fernandez *et al.* [32] integrated the DDA mechanism with Rhythm-Group Theory in a 2D platform exergame, where the exergame difficulty level was adjusted in real-time based on the user’s attention level. Chanel *et al.* [33] proposed the use of deep data-driven affective models capable of detecting anxiety and boredom from the user’s physiological signals to adjust the gaming difficulty and increase user engagement.

The adaptation of the game timer represents a prevalent and common implemented mechanism within DDA. Adjusting the timer’s speed and duration during exergame gameplay holds substantial potential to influence the user’s overall experience [34]. Typically, Galy *et al.* [35] observed that introducing time pressure and modifying task difficulty levels were an effective strategy for enhancing user motivation. Another principal adaptation strategy involves adjusting the in-game virtual object’s speed, although this proves more effective in 2D games compared to more intricate 3D games, possibly due to the simpler design nature of 2D games [28], [33].

Procedural Content Generation (PCG) serves as an alternative method, wherein the level design is procedurally adjusted to either increase or decrease the game difficulty level. Huber *et al.* [36] implemented PCG for DDA in IVR exergaming, dynamically generating game levels tailored to users’ performances and abilities. This approach aims to alleviate monotony over extended periods and enable concurrent adjustments of both cognitive and physical demands during IVR exergaming. Furthermore, research highlights the positive effects of other DDA mechanisms, such as the implementation of varying task complexity [37], in-game feedback [38], and hint-supporting mechanisms [27].

Despite the promising outcomes of DDA in motivating users, there exists a gap in comprehensive research analyzing changes in users’ emotional states concerning the integration of DDA in IVR exergaming. This gap is particularly notable concerning the suitability and comfort of the DDA mecha-

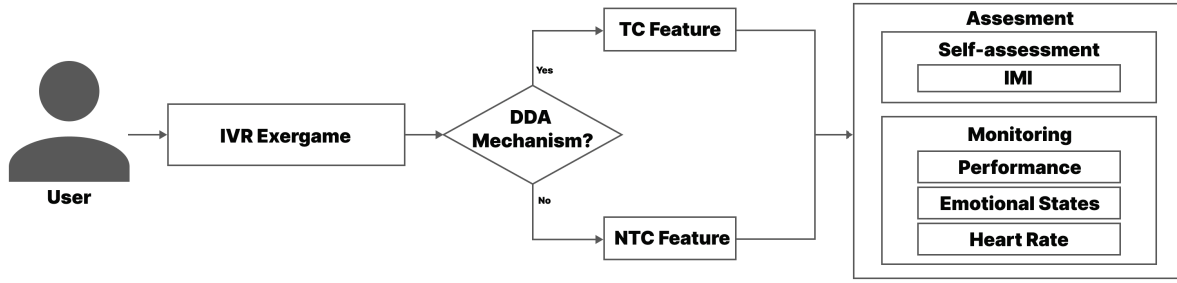


Fig. 1: System overview.



(a)



(b)

Fig. 2: The proposed design of the (a) IVR soccer exergaming and (b) illustration of a participant engaged in the IVR exergaming, utilizing virtual hands through the Meta Quest Pro device.

nism for exergaming users, with a focus on automating the adjustment of game difficulty levels based on users' physical performance, such as their gaming score.

III. METHODS

Fig. 1 illustrates the system overview of the IVR exergaming design, including the proposed TC and NTC feature-based DDA mechanism, followed by an assessment of the emotional impact on DDA in IVR exergaming.

A. Design of IVR Exergaming

To conduct the study, a self-designed IVR soccer exergame was developed utilizing Unity3D [39] and integrating the Meta Movement SDK [40], which enables facial feature extraction during gameplay. In this game, the user assumes the role of a goalkeeper positioned in front of a virtual soccer goal. The objective is to prevent soccer balls from entering the goal by using virtual hands to block them as illustrated in Fig. 2a. Each successful defense resulted in the user earning one point.

In this study, the TC feature was used as a DDA mechanism where prior research has underscored TC's efficacy in en-

hancing user motivation throughout exergaming activities [21], [34]. The design and integration of TC within this research drew upon the framework established by [41], with minor refinements tailored to the specifics of this inquiry. The TC adaptation is activated every 10 seconds during gameplay. The TC feature underwent manipulation through a time speed multiplier M , determined based on the gaming score GS and the real-time standard gaming score GS_{sc} , as outlined below:

$$M(GS, GS_{sc}) = \begin{cases} 2.2, & \text{if } GS > GS_{sc} + 6 \\ 1.8, & \text{if } GS_{sc} + 3 < S \leq GS_{sc} + 6 \\ 1.4, & \text{if } GS_{sc} < S \leq GS_{sc} + 3 \\ 1.0, & \text{if } GS = GS_{sc} \\ \frac{1}{1.4}, & \text{if } GS_{sc} - 3 \leq GS < GS_{sc} \\ \frac{1}{1.8}, & \text{if } GS_{sc} - 6 \leq GS < GS_{sc} - 3 \\ \frac{1}{2.2}, & \text{if } GS < GS_{sc} - 6 \end{cases} \quad (1)$$

To clarify, the GS_{sc} value increments by one each second. At the start of a game, with an initial M value of 1.0, if a user's GS value exceeds 15 after the first 15 seconds, the M value is

adjusted based on both GS and GS_{sc} values. Conversely, if the GS value drops below 15, the M value decreases accordingly. However, if the GS value reaches precisely 15 after the 15 seconds, the M value remains at 1.0.

B. Experiment Design

A total of 14 participants, with an average age of 25.4 ± 2.35 years, comprising 10 males and 5 females, were enlisted from the university to participate in this study. Prior to the initiation of the experiments, researchers conducted a detailed briefing to inform participants with the study's objectives. After obtaining their agreement, participants were required to provide signed written consent. Subsequently, a trial session was conducted to familiarize participants with the Meta Quest Pro VR device's operation and virtual interactions within the game.

Participants were instructed to participate in the same game twice, once with the TC feature and once without (Non-TC or NTC). The order of these sessions was randomized, with half of the participants commencing with TC-based IVR exergaming followed by NTC-based IVR exergaming, and the other half initiating with NTC-based IVR exergaming. The total duration for NTC-based IVR exergaming was fixed at 300 seconds, while the duration for TC-based IVR exergaming varied based on participants' gaming performance. The study underwent a comprehensive review and obtained approval from the university's faculty research ethics committee, adhering to ethical review procedures outlined in the university's research conduct and research ethics guidelines [42].

Throughout the experiments, participants wore a Polar HR armband [43] to record HR data at a sampling rate of 1 Hz. Eyebrow movement (EM) and facial action unit data [44] were collected at a sampling rate of 10 Hz, starting 15 seconds after the game's commencement. Upon completing the experiments, each participant completed the IMI (see Section III-C).

C. IMI

The IMI serves as a multidimensional measurement tool designed to evaluate participants' subjective experiences concerning intrinsic motivation within the context of the experiments conducted in this study [16]. The standard IMI employed in this study comprises 22 scales relating to intrinsic motivation, which are further categorized into four sub-scales: enjoyment/interest, perceived choice, perceived competence, and pressure/tension. The enjoyment/interest sub-scale functions as a self-report measure of intrinsic motivation, while perceived choice and perceived competence are theorized to serve as positive predictors for both self-report and behavioral measures of intrinsic motivation. Conversely, pressure/tension is theorized to act as a negative predictor of intrinsic motivation.

D. Data Analysis

The emotional state of each participant was assessed using the Facial Action Coding System [44], which covers a comprehensive set of seven distinct emotions: "happy", "sad", "surprise", "fear", "anger", "disgust", and "contempt".

It is noteworthy that this study excluded the emotions of "surprise", "fear", "anger", "disgust", and "contempt" since the designed exergame was not specifically tailored to detect these emotions. A paired samples t -test was employed to evaluate the differences in emotional states, including heart rate (HR), eyebrow movement (EM), IMI, and gaming scores (GS), between conditions with TC and NTC. Additionally, the correlation between emotional state and HR, as well as between emotional state and EM, was computed and analyzed using Spearman's correlation test. Statistical Product and Service Solutions (SPSS) [45] was utilized as the statistical analysis tool.

IV. RESULTS AND DISCUSSION

The experimental findings revealed that the TC-based IVR exergaming was completed in a minimum duration of 143 seconds. Therefore, a sample size of 143 seconds was employed for the analysis of TC-related data. The mean emotional state μ was computed by considering the counts of detected emotions during gameplay, and the results are outlined in Table I.

The findings revealed that participants reported higher levels of happiness in the NTC condition ($\mu_{NTC}^{ES} = 4.71$, $STD_{NTC}^{ES} = 1.07$) compared to the TC condition ($\mu_{TC}^{ES} = 2.93$, $STD_{TC}^{ES} = 1.21$), with a $t = 3.617$ and $p = 0.003$. A greater quantity of sadness was observed in the NTC condition ($\mu_{NTC}^{ES} = 1.64$, $STD_{NTC}^{ES} = 1.22$) than in the TC condition ($\mu_{TC}^{ES} = 1.50$, $STD_{TC}^{ES} = 0.94$), with a $t = 0.322$ and $p = 0.752$. Meanwhile, neutral emotion was less prevalent in the NTC condition ($\mu_{NTC}^{ES} = 7.50$, $STD_{NTC}^{ES} = 1.74$) compared to the TC condition ($\mu_{TC}^{ES} = 9.64$, $STD_{TC}^{ES} = 1.45$), and this difference was statistically significant with a $t = -3.238$ and $p = 0.007$.

Overall, participants exhibited a greater range of happiness and sadness emotions in the NTC conditions, implying increased variability in emotional behavioral changes. Conversely, participants in the TC conditions predominantly displayed neutral emotions, indicating greater stability in emotional changes and suggesting a high adaptation of the TC feature in IVR exergaming.

However, upon thorough result analysis, no significant difference was observed in the changes of the EM data. Specifically, the EM values were lower in the NTC condition ($\mu_{NTC}^{EM} = 0.022$, $STD_{NTC}^{EM} = 0.038$) compared to the TC condition ($\mu_{TC}^{EM} = 0.025$, $STD_{TC}^{EM} = 0.035$), showing low significance for both TC and NTC conditions with a $t = -0.696$ and a $p = 0.487$. Conversely, a significant difference was found in the changes of HR, which exhibited lower values in the NTC condition ($\mu_{NTC}^{HR} = 107.73$, $STD_{NTC}^{HR} = 13.03$) than in the TC condition ($\mu_{TC}^{HR} = 110.06$, $STD_{TC}^{HR} = 12.58$), displaying significant differences for both TC and NTC conditions with a $t = -10.939$ and $p < 0.001$.

On the other hand, the mean GS per second in the NTC condition ($\mu_{NTC}^{GS} = 1.492$, $STD_{NTC}^{GS} = 0.197$) is notably lower than in the TC condition ($\mu_{TC}^{GS} = 1.520$, $STD_{TC}^{GS} = 0.232$), displaying significant differences for both TC and NTC conditions with a $t = -0.342$ and $p = 0.736$. In summary,

TABLE I: Statistical results of changes in emotional state (ES) under NTC and TC conditions

Emotion	μ_{NTC}	STD_{NTC}	μ_{TC}	STD_{TC}	t -value	p -value
Happy	4.71	1.07	2.93	1.21	3.617	0.003
Sad	1.64	1.22	1.50	0.94	0.322	0.752
Neutral	7.50	1.74	9.64	1.45	-3.238	0.007

TABLE II: Statistical results of changes in EM, HR and GS data under NTC and TC conditions

Data	μ_{NTC}	STD_{NTC}	μ_{TC}	STD_{TC}	t -value	p -value
EM	0.022	0.038	0.025	0.035	-0.696	0.487
HR	107.73	13.03	110.06	12.58	-10.939	< 0.001
GS	1.492	0.197	1.520	0.232	-0.342	0.736

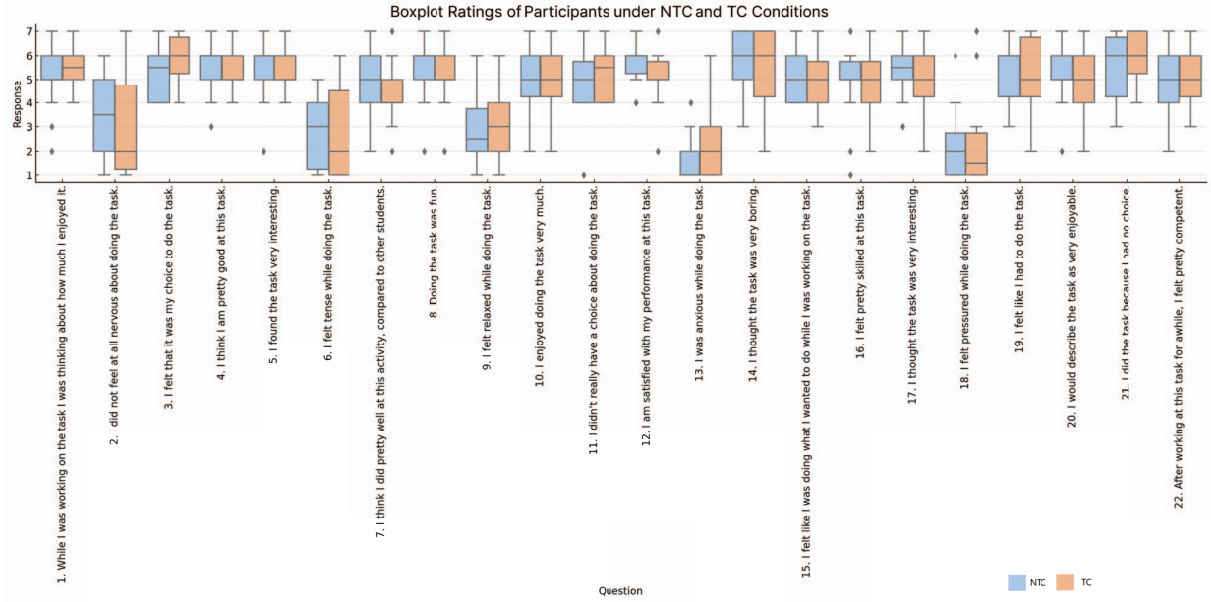


Fig. 3: Statistic results of IMI under NTC and TC conditions.

the results suggest that participants achieved better physical performance in the TC conditions, indicating the effectiveness of the individually tailored difficulty level. Table II provides a summary of the mean values of the changes in the EM, HR and GS data.

Meanwhile, a weak negative correlation between ES and EM data was identified for participants under the NTC condition, with a Spearman's correlation coefficient value of $c = -0.172$ and $p = 0.016$. Similarly, a weak negative correlation between ES and EM data was also observed for participants under the TC condition, with $c = -0.027$ and $p = 0.706$. Meanwhile, there was a negative correlation between ES and HR data for both TC ($c = -0.061$, $p = 0.396$) and NTC ($c = -0.074$, $p = 0.300$) conditions.

On the other hand, no significant difference is identified in the IMI for participant under the NTC and TC conditions ($t = 0.206$, $p = 0.837$), as shown in Fig. 3. Specifically, participants experienced game under NTC condition presented the mean rating of the enjoyment as $\mu_{NTC}^{EJ} = 5.34$ and $STD_{NTC}^{EJ} = 0.20$, perceived competence as $\mu_{NTC}^{PC} = 5.17$ and $STD_{NTC}^{PC} = 0.42$, perceived choice as $\mu_{NTC}^{PE} = 5.23$ and $STD_{NTC}^{PE} =$

0.26 , and tension as $\mu_{NTC}^{TS} = 2.60$ and $STD_{NTC}^{TS} = 0.69$.

Meanwhile, participants who experienced the game with the TC feature reported mean ratings on the IMI in the aspect of enjoyment as $\mu_{TC}^{EJ} = 5.27$ with $STD_{TC}^{EJ} = 0.17$, perceived competence as $\mu_{TC}^{PC} = 4.94$ with $STD_{TC}^{PC} = 0.37$, perceived choice as $\mu_{TC}^{PE} = 5.44$ with $STD_{TC}^{PE} = 0.44$, and tension as $\mu_{TC}^{TS} = 2.64$ with $STD_{TC}^{TS} = 0.42$. Overall, the results revealed that mean ratings for enjoyment and perceived competence were higher under the NTC condition compared to the TC condition, while the mean ratings for perceived choice and tension were higher under the TC condition.

The study results revealed that adjusting the timer based on users' GS had a discernible impact on their emotional responses. The primary determinant of users' emotions in this context was the level of challenge presented by the exergame. In the NTC condition, where difficulty levels were not customized according to users' exergaming ability, participants exhibited a diverse range of emotional experiences. In contrast, TC-based gaming provided users with a more comfortable exergaming experience, associated with fewer extreme positive or negative emotions. The findings also corroborated the

principles of flow theory [46], indicating that optimizing the exergame challenge level within a comfort zone could sustain user engagement without causing frustration or boredom.

Throughout the study, it was observed that as the timer speed increased with improved GS, participants tended to achieve even higher GS. Consequently, during TC adaptation intervals, participants felt compelled to enhance their movement speed, resulting in elevated HRs due to increased focus. This phenomenon aligns with the finding of [47], which elucidated that heightened HR correlates with increased oxygenated blood circulation during periods of intensified physical activity. Conversely, participants with more experience in IVR gaming achieved high GS in less time, but their overall experience remained consistent when exposed to TC-based exergaming. In contrast, less experienced IVR exergaming participants performed better with a slower timer change, exhibiting more comfortable gaming experiences.

Moreover, during the NTC-based exergaming experiments, it was noted that participants who perceived their performances as below expectations tended to experience a sense of diminished control over the game progress. This resulted in increased emotional fluctuations and a decline in gaming performance. In conclusion, this study proposed that neutral emotion serves as a promising satisfactory indicator for the design of the DDA mechanism in IVR gaming.

V. CONCLUSION

This study examined changes in participants' emotional behavior during IVR exergaming under two conditions: games with the TC feature and without the TC feature (NTC). The results revealed that participants in the exergame with the TC feature, a popular DDA mechanism, exhibited a higher frequency of neutral emotions and improved in-game performance. Conversely, participants in the exergame without the TC feature displayed more variability in emotions, suggesting that the difficulty level provided diverse experiences, lacking personalized adjustments for individual needs. Moreover, the DDA mechanism with the TC feature in IVR exergaming, providing personalized challenge adjustments, could lead to more neutral emotions. These preliminary findings lay the groundwork for future research into the development of autonomous DDA mechanism-enabled IVR exergaming, which takes into account changes in users' physical and mental states.

The study has certain limitations, such as a relatively small sample size, potentially limiting the generalizability of the results. Additionally, exploring different exergames with participants from diverse demographics is necessary for a comprehensive understanding of emotional behavior responses to DDA. Future research should consider investigating various DDA mechanisms beyond TC and aim to design an optimized autonomous DDA mechanism based on both the physical and mental conditions of IVR exergaming users.

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